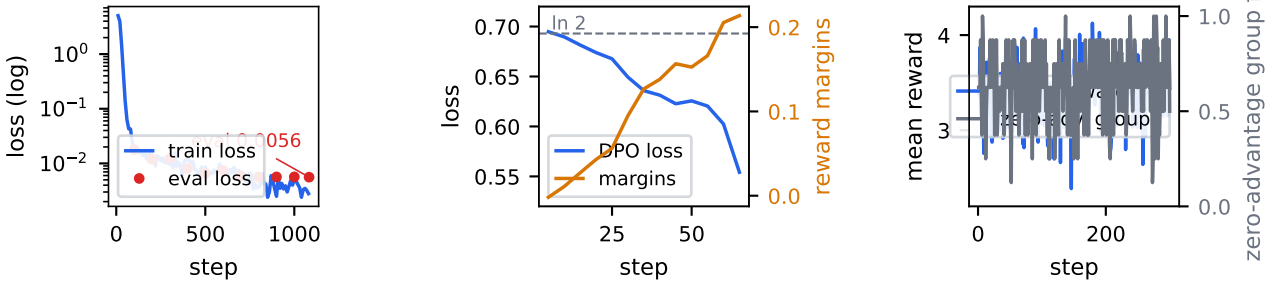
(a) SFT
loss (log)
10^0
10^{-1}
10^{-2}
train loss
eval loss
0.0056
0
500
1000
step
(b) DPO
ln 2
0.70
0.65
0.60
0.55
loss
DPO loss
margins
0.2
0.1
0.0
reward margins
25
50
step
(c) GRPO
mean reward
4
3
1.0
0.5
0.0
zero-advantage group frac.
0
200
step